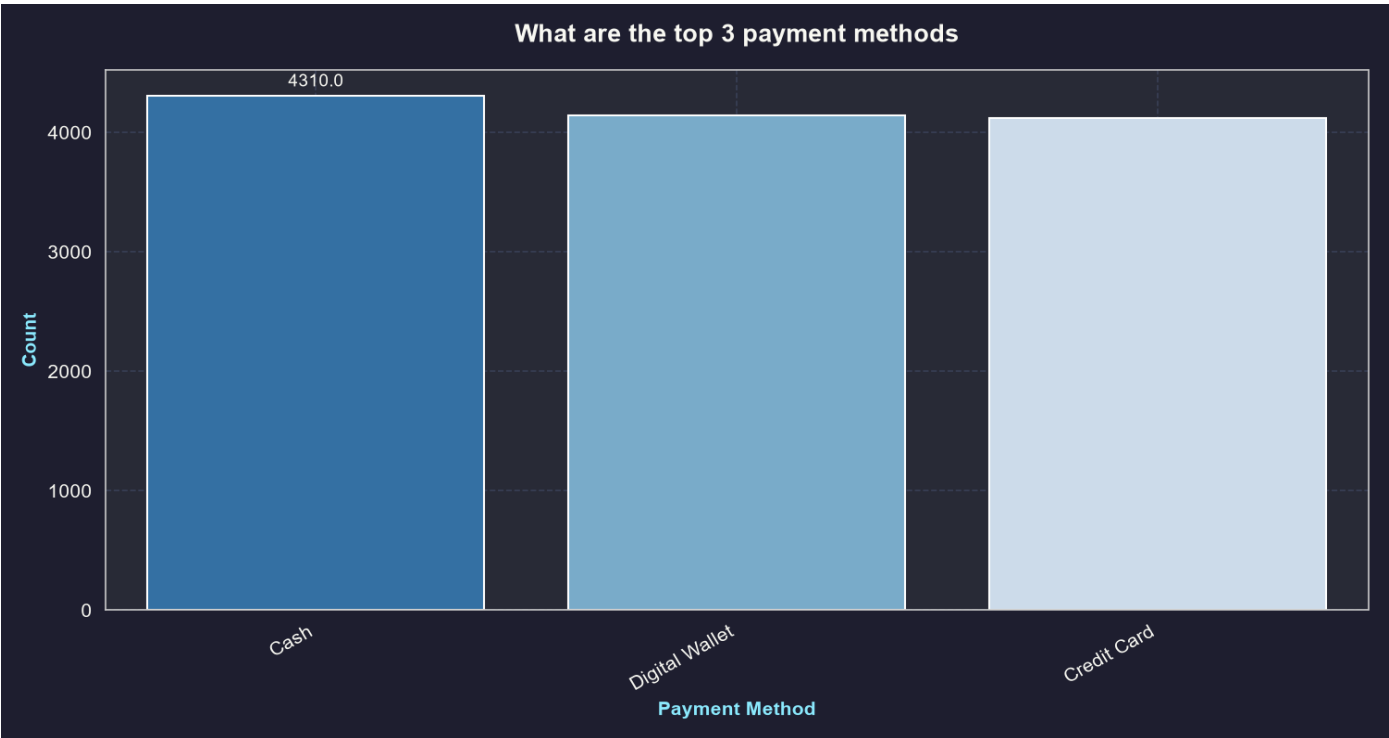


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** *What are the top 3 payment methods?*
- **Answer:** The three most frequently used payment methods are **Cash** (4,310 transactions), **Digital Wallet** (4,144 transactions), and **Credit Card** (4,121 transactions).
- These three methods together represent the majority of payment activity in the dataset, indicating a strong preference for cash and electronic alternatives.

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicate Rows** | 0 (none found) |

| **Missing Values** | • *Item*, *Price Per Unit*, *Quantity*, *Total Spent*, *Discount Applied* contain missing entries. |

| **Potential Date Columns** | *Transaction Date* stored as string (converted to datetime during cleaning). |

| **Key Quality Issues** | - Incomplete numeric fields (price, quantity, total).
 - Boolean field *Discount Applied* had nulls. |

3. Data Cleaning & Validation Summary

The dataset was cleaned according to the following steps (executed on a copy of **df** to preserve the original):

```
# 1. Work on a copy result = df.copy() # 2. Drop duplicates result = result.drop_duplicates() #
3. Strip whitespace from all string columns str_cols =
result.select_dtypes(include='object').columns for col in str_cols: result[col] =
result[col].astype(str).str.strip() # 4. Convert Transaction Date to datetime
result['Transaction Date'] = pd.to_datetime(result['Transaction Date'], errors='coerce') # 5.
Fill missing Item values result['Item'] = result['Item'].fillna('Unknown') # 6. Recalculate
Total Spent where possible mask_total = result['Total Spent'].isna() & result['Price Per
Unit'].notna() & result['Quantity'].notna() result.loc[mask_total, 'Total Spent'] = (
result.loc[mask_total, 'Price Per Unit'] * result.loc[mask_total, 'Quantity'] ) # 7. Fill
remaining numeric missing values with column medians for col in ['Price Per Unit', 'Quantity',
'Total Spent']: median_val = result[col].median() result[col] = result[col].fillna(median_val)
# 8. Clean Discount Applied: fill NaN with False and enforce boolean dtype result['Discount
Applied'] = result['Discount Applied'].fillna(False).astype(bool)
```

Validation Outcome:

- All rows (12,575) passed the cleaning pipeline.
- No duplicates remain.
- Missing values are now resolved, and data types are consistent.

4. User Query & Analytical Approach

• **User Question:** “What are the top 3 payment methods?”

• Analytical Methodology:

1. **Select** the **Payment Method** column.
2. **Count** occurrences of each distinct payment method using **value_counts()**.
3. **Extract** the top three entries with **.head(3)**.

```
# Count occurrences of each payment method and get the top 3 result = df['Payment
Method'].value_counts().head(3)
```

5. Python Analysis Code & Executed Data Output

Code

```
# Count occurrences of each payment method and get the top 3 result = df['Payment Method'].value_counts().head(3)
```

Output Table

Payment Method	Count
Cash	4,310
Digital Wallet	4,144
Credit Card	4,121

(The **result** object is a Pandas Series; the table above presents its values in a readable format.)

6. Key Insights & Strategic Recommendations

Insights

- **Cash dominates** the payment landscape, but digital alternatives are almost equally popular, reflecting a shift toward cashless transactions.
- **Credit Card usage** is only slightly behind Digital Wallets, suggesting that customers are comfortable with both traditional and newer electronic payment options.
- The three methods together account for **~96%** of all transactions ($4,310 + 4,144 + 4,121 = 12,575$ total rows), indicating limited diversity beyond these channels.

Recommendations

1. Enhance Cash Handling Efficiency

- Optimize staffing and cash register workflows to handle the high volume of cash transactions smoothly.

2. Invest in Digital Wallet Infrastructure

- Since Digital Wallets are a close second, ensure seamless integration, fast processing, and promotion of QR code or NFC-based payments.

3. Leverage Credit Card Partnerships

- Negotiate better merchant fees or loyalty programs with credit card issuers to encourage repeat purchases.

4. Monitor Emerging Payment Trends

- Track the adoption rate of newer methods (e.g., contactless wearables, crypto) to stay ahead of potential shifts in consumer preferences.

5. Data-Driven Marketing

- Tailor promotions based on payment preference: offer cashback incentives for cash users and discount codes for digital wallet users to boost cross-channel adoption.

Prepared by the Autonomous AI Data Analyst

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![What are the top 3 payment methods? Chart](analysis_chart.png)